

Lab6

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May 2026

1 Overview

Static Random-Access memory (SRAM) is a memory that store data. The size of the individual, group, or transistor directly affects the reading and writing of SRAM's. Therefore, the objective of this report is to walk through sizing sweeping to led to a better understanding of how sizing affects these functionality. In 6T SRAM, there is three group of transistors including access transistors, pull up and pull down of inverter. During read operation, access transistor directly affect how fast current discharge during read. This could result in reading upset when access transistor is stronger than that of pull down network. Pull down network of coupled inverter also affect when it is weaker comparing to access transistor. During write operation, both pull up network and access transistor can cause writing failure when pull up network is significantly stronger than that of access network. There is a trade off between the need for weaker access network for writing and strong access network during the reading. In order to isolate the requirement, 8T SRAM with 2T leads to a separate branch of reading. Thus, access transistor can be optimized writing without compromising the read requirement.

Part	Device Name	Size(W/L)
6T SRAM Read 0 Failure	M6	32 (W=1.6u, L = 50n)
6T SRAM Read 0 Failure	M3	2 (W=100n, L = 50n)
6T SRAM Write 0 Failure	M6	1.8(W=90n, L = 50n)
6T SRAM Write 0 Failure	M4	5.75(W=285n, L = 50n)
8T SRAM Write Failure	M2	1.7(W = 85n, L = 50n)
8T SRAM Write Failure	M3	5.75(W=287.5n, L = 50n)

Figure 1: Enter Caption

2 Simulation conditions and sizing strategy

2.1 6T SRAM

For initial ratio, access transistor has $W/L_{min} = 2$ with $L_{min} = 50n$. The PMOS and NMOS of the coupled inverter has the W/L_{min} ratio = 5.75 and 2, respectively. During reading, data = 0 and dataB = 1V. Both BL and BLB precharged to VDD. Then WL is set to VDD for reading. When WL pulled up, mismatch supposed to discharge, the BL. However, as discharging. If access transistor is much stronger than pull-down transistor. Voltage will build up and can cause a read upset. Therefore, this report construct a testbench sweeping ratio of access transistor with $L = L_{min} = 50n$, and Width of access transistor= ratio* L_{min} . The remaining transistor use the initial ratio. When finding the size for read fail, ratio of W/L above 32 is show to see a read upset.

During testing for writing, data is = 1 and dataB is = 0. BL has 0 loaded in and BLB has 1 loaded in using DC supply. The transistors set up are the same and we continue to sweep through access ratio. When WL is pulled up, the ratio is swept to see the ratio before the data failed to be written in 0. The ratio W/L for access data = 1.7 or less would cause a fail in writing 0.

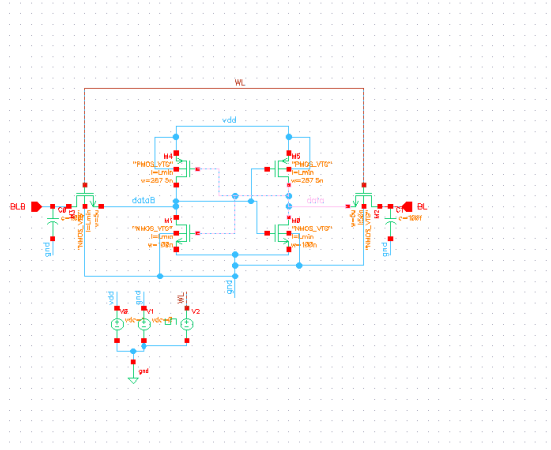
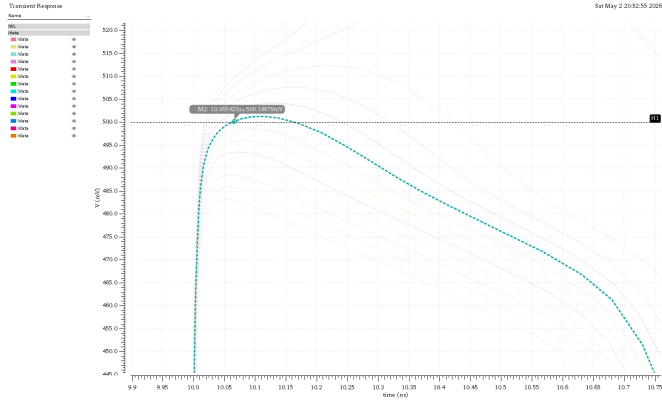
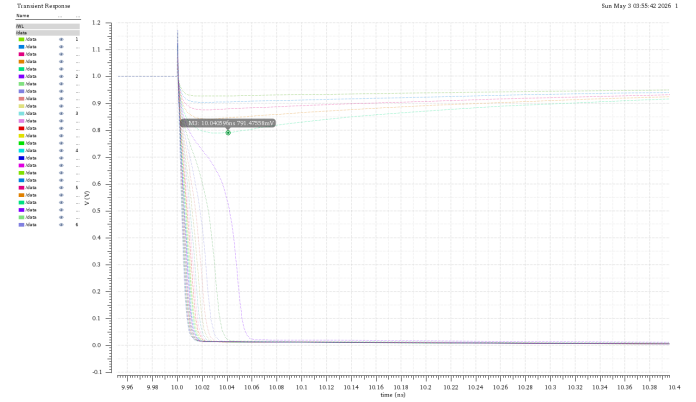


Figure 2: 6T SRAM schematics

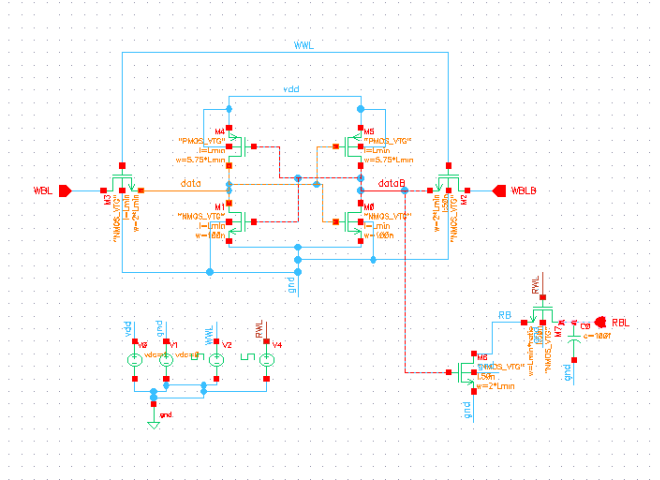


(a) Sweeping for read fail ratio

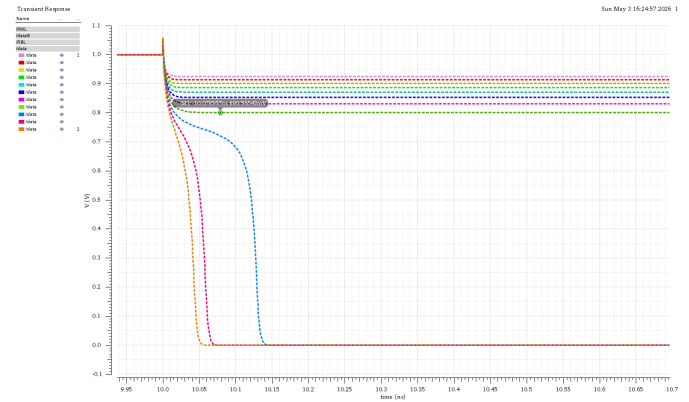


(b) Sweeping for write fail ratio

2.2 8T SRAM



(a) 8T SRAM schematics



(b) Sweeping for write fail ratio

For 8T sizing, the ratio is done similar to the initial sizing with additional M7 and M8 with $W/L = 2$. In this case, read is not affected by any sizing due to the data is the input at a gate. Unless written data or stored data is less than the threshold voltage, other than that, the data would be correctly read without the issue of reading offset. In the case of writing, the writing fails is still affect by access transistor with $\text{ratio} * L_{\min}$ and ratio swept across different ratio. W/L for access transistor are found to fail when access transistor is below 46.

3 Results and Discussion

The results are as predicted as writing structure is similar, which yield the similar writing fail ratio, while the reading is shown more stable. 8T is shown to have a better read stability where it is less susceptible to read upset. This is because data is used to turn on or off another transistor, isolating data and bit line. Bitline. 6T is shown to have a slightly write stability where transistor 1.7 W/L can still read correctly. However, writing stability are similar between 6T and 8T.